

BUCURESTI FILARET, RomaniaWMO: **154220**Lat: **44.4122N**Lon: **26.0939E**Elev: **84**StdP: **100.33**Time Zone: **2.00 (EUE)**Period: **86-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.9	-7.5	-12.4	1.3	-8.2	-10.4	1.6	-6.3	12.5	-0.3	11.6	-0.4	1.5	220	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.7	34.8	21.8	33.3	21.1	31.7	20.6	23.5	31.3	22.6	30.2	21.8	29.2	1.1	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	16.3	25.8	20.4	15.2	25.1	19.5	14.4	24.5	70.4	31.1	67.0	30.2	64.0	29.4	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	6.1	4.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.9	-0.7	3.8	8.1	12.5	18.0	22.9	24.2	24.7	19.6	12.8	6.2	2.2
	DBStd	9.43	4.36	4.46	4.41	4.28	3.09	2.94	2.49	2.96	3.68	4.28	4.42	3.59
	HDD10.0	1019	332	177	88	25	0	0	0	0	0	23	131	243
	HDD18.3	2611	591	406	318	179	43	3	0	1	30	174	365	501
	CDD10.0	2077	0	4	28	99	249	386	440	457	288	111	16	0
	CDD18.3	628	0	0	0	3	33	139	182	199	69	3	0	0
	CDH23.3	5977	0	1	4	59	317	1252	1684	2016	596	48	1	0
	CDH26.7	2505	0	0	0	9	68	495	727	988	211	8	0	0
Wind	WSAvg	1.4	1.7	1.7	1.9	1.6	1.4	1.2	1.1	1.1	1.1	1.3	1.4	1.7
Precipitation	PrecAvg	611	43	40	39	46	71	80	57	58	44	32	53	47
	PrecMax	854	132	108	128	96	260	178	134	176	155	143	207	185
	PrecMin	412	5	8	2	5	10	15	8	3	1	0	1	1
	PrecStd	108	32	27	33	23	51	42	33	41	41	27	42	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.5	19.0	23.6	27.9	30.6	35.8	35.8	37.0	33.6	27.5	21.4	13.4
		MCWB	8.0	12.5	13.9	17.4	19.3	23.2	22.5	21.9	20.1	17.8	14.7	9.1
	2%	DB	8.6	14.8	20.3	25.0	28.5	33.4	34.2	35.1	31.3	24.2	17.1	10.9
		MCWB	6.1	10.1	12.1	16.2	18.9	22.5	21.6	21.6	19.3	16.1	12.9	8.2
	5%	DB	6.3	12.2	17.8	22.2	26.9	31.2	32.6	33.7	28.8	21.8	14.3	8.6
		MCWB	4.8	8.5	10.9	14.4	18.1	21.7	21.1	20.8	18.7	15.0	11.3	6.6
	10%	DB	4.6	9.9	15.3	20.3	24.9	29.5	30.7	32.1	26.6	19.5	12.0	6.7
		MCWB	3.5	7.1	9.7	13.5	16.8	20.9	20.5	20.5	18.0	14.1	10.4	5.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.5	13.1	14.5	18.0	20.9	25.2	24.2	24.7	22.1	18.4	15.2	10.2
		MCDB	11.2	18.3	21.8	26.5	27.6	33.3	31.5	32.2	30.2	24.7	20.1	12.2
	2%	WB	6.4	10.8	12.6	16.7	19.9	23.6	23.1	23.2	20.5	17.0	13.6	8.3
		MCDB	8.1	14.7	19.0	24.1	26.5	31.0	30.8	31.5	27.5	21.9	17.0	10.6
	5%	WB	5.1	8.9	11.3	15.4	18.9	22.5	22.3	22.2	19.5	15.9	12.2	6.9
		MCDB	6.3	11.4	16.8	21.4	25.3	29.5	29.6	30.5	26.6	20.6	14.1	8.4
	10%	WB	3.6	7.5	10.0	14.0	17.8	21.6	21.6	21.4	18.6	14.6	10.7	5.6
		MCDB	4.6	9.4	14.6	18.8	23.7	28.2	28.4	29.7	25.4	18.5	12.1	6.8
Mean Daily Temperature Range	5% DB	MDBR	6.1	8.0	9.7	10.9	11.5	11.2	11.7	12.7	11.7	10.1	6.5	5.7
		MCDBR	8.4	12.0	14.5	15.0	14.2	13.6	14.3	14.8	14.8	14.2	10.9	9.0
		MCWBR	5.7	7.5	7.4	7.6	6.1	5.4	4.9	4.8	5.5	7.0	6.5	6.1
	5% WB	MCDBR	7.6	10.6	13.3	14.2	12.9	12.8	12.4	13.4	12.9	12.9	9.3	9.4
		MCWBR	5.4	6.8	7.2	7.3	5.9	5.6	4.9	4.9	5.2	6.7	5.8	6.2
Clear-Sky Solar Irradiance	taub	0.349	0.381	0.416	0.451	0.426	0.436	0.446	0.460	0.424	0.403	0.376	0.352	
	taud	2.268	2.186	2.146	2.078	2.226	2.233	2.213	2.167	2.233	2.261	2.286	2.281	
	Ebn at Noon	744	784	808	811	847	838	823	795	792	752	705	696	
	Edh at Noon	87	112	133	154	137	137	138	140	121	103	85	78	
All-Sky Solar Radiation	RadAvg	1.38	2.21	3.43	4.71	5.87	6.35	6.43	5.74	4.16	2.70	1.49	1.14	
	RadStd	0.21	0.34	0.37	0.48	0.44	0.44	0.42	0.32	0.32	0.32	0.20	0.17	